

60-DAY DATA SCIENCE ROADMAP

A Structured Learning Plan from Python Fundamentals to Machine Learning

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Python | Data Analysis | Statistics | Machine Learning | Deep Learning | Portfolio

ROADMAP OVERVIEW

This roadmap provides a structured, practical sequence for learning core Data Science skills. It moves from programming fundamentals to data analysis, statistical reasoning, machine learning, introductory deep learning, and a final portfolio project.

Suggested daily study routine

- Review the previous lesson and key terms.
- Study the new topic and take concise notes.
- Run the code examples and rewrite them independently.
- Complete practice tasks before reviewing solutions.
- Save code, notes, and project progress in GitHub.

How to use this roadmap

Treat the daily schedule as a guide, not a deadline. Revisit difficult topics, test your understanding with small experiments, and prioritize clear reasoning and reproducible work over speed.

PHASE 1 | Python Foundations | Days 1–10

Day 01	Setup, program structure, print(), comments, variables and naming rules.
Day 02	Data types, type checking and conversion, input(), output formatting.
Day 03	Arithmetic, comparison and logical operators, Boolean values, precedence.
Day 04	Conditionals, loops, break/continue, nested logic and functions.
Day 05	Lists, indexing, slicing, list methods and iteration.
Day 06	Tuples, sets, dictionaries, membership and nested structures.
Day 07	File handling, paths, text files, CSV and JSON.
Day 08	Exceptions, try/except, debugging, tracebacks and input validation.
Day 09	Modules, packages, imports, pip and virtual environments.
Day 10	Object-oriented programming: classes, objects, attributes, methods and constructors.

PHASE 2 | Numerical Computing and Data Analysis | Days 11–16

Day 11	NumPy arrays, dimensions, shape, dtype, indexing and vectorized operations.
Day 12	Broadcasting, aggregation, masks, sorting, random sampling and reshaping.
Day 13	Pandas Series and DataFrames, reading, inspecting, selecting and filtering.
Day 14	Data cleaning, missing values, duplicates, data types, strings and outliers.
Day 15	Matplotlib: line, bar, scatter and histogram plots, labels, legends and subplots.
Day 16	Seaborn: distributions, categorical plots, heatmaps and pair plots.

PHASE 3 | Statistics and Probability | Days 17–21

Day 17	Descriptive statistics: mean, median, mode, variance, standard deviation, quartiles and IQR.
Day 18	Probability, sample spaces, events, complements, conditional probability and independence.
Day 19	Probability distributions: PMF, PDF, CDF, Bernoulli, binomial, normal and uniform.
Day 20	Sampling, bias, sampling distributions, standard error, Central Limit Theorem and confidence intervals.
Day 21	Hypothesis testing, p-values, significance levels, one-/two-tailed tests and t-tests.

PHASE 4 | Exploratory Data Analysis | Days 22–28

Day 22	Correlation and covariance, Pearson vs. Spearman, heatmaps and correlation limitations.
Day 23	Bivariate analysis and relationships between numerical and categorical variables.
Day 24	EDA workflow: questions, schema, distributions, missing data and outliers.
Day 25	Feature relationships, confounding, target leakage and spurious relationships.
Day 26	Data storytelling, chart selection, annotations and communicating limitations.
Day 27	EDA mini-project: clean, explore, visualize and summarize a real dataset.
Day 28	Refine the EDA notebook, document findings and make the work reproducible.

PHASE 5 | Data Preparation and SQL | Days 29–35

Day 29	Data types, date/time handling, parsing and validation.
Day 30	Categorical encoding: one-hot and ordinal encoding.
Day 31	Feature scaling: standardization and min-max scaling.
Day 32	Train/validation/test splits, random states, stratification and data leakage.
Day 33	SQL: SELECT, WHERE, ORDER BY, LIMIT, DISTINCT and NULL handling.
Day 34	SQL aggregation and joins: GROUP BY, HAVING, COUNT, SUM, AVG, INNER/LEFT JOIN.
Day 35	Data preparation project: combine, clean, validate and prepare data for modeling.

PHASE 6 | Machine Learning Fundamentals | Days 36–45

Day 36	Machine learning overview: supervised/unsupervised learning, features, targets and workflow.
Day 37	Linear regression, coefficients, residuals, MAE, MSE, RMSE and R^2 .
Day 38	Multiple regression, baseline models, evaluation and coefficient interpretation.
Day 39	Logistic regression, predicted probabilities, thresholds and confusion matrices.
Day 40	Classification metrics: accuracy, precision, recall, F1, ROC-AUC and class imbalance.
Day 41	Decision trees, splitting, depth, overfitting and limits of feature importance.
Day 42	Random forests, ensemble learning, bagging and tuning basics.
Day 43	K-nearest neighbors, distance measures, scaling and selecting k.
Day 44	K-means clustering, feature scaling, choosing k and interpreting clusters.
Day 45	Model comparison, cross-validation, pipelines and error analysis.

PHASE 7 | Improving Models | Days 46–52

Day 46	Bias-variance trade-off, underfitting, overfitting and regularization.
Day 47	Hyperparameter tuning using grid search, randomized search and cross-validation.
Day 48	Feature engineering, domain features and leakage checks.
Day 49	Missing-data strategies and scikit-learn preprocessing pipelines.
Day 50	Imbalanced classification, class weights, resampling concepts and thresholds.
Day 51	Model interpretation: coefficients, permutation importance and limitations.
Day 52	End-to-end supervised learning project with clear evaluation.

PHASE 8 | Introductory Deep Learning and Portfolio | Days 53–60

Day 53	Neural network intuition: neurons, layers, weights and activation functions.
Day 54	Tensors, forward pass, loss functions, gradient descent and backpropagation.
Day 55	Build a small neural network with a suitable beginner-friendly framework.
Day 56	Training loops, epochs, batch size, learning rate and validation loss.
Day 57	Overfitting, dropout, regularization, evaluation and reproducibility.
Day 58	Capstone project: define a question, source data responsibly, clean, explore and model.
Day 59	Polish GitHub portfolio: notebook, requirements, data sources, README, plots and limitations.
Day 60	Review the complete workflow, present the capstone and plan the next learning stage.

CORE TOOLS

- Python for programming and analysis.
- Jupyter Notebook or Google Colab for interactive experiments.
- NumPy and Pandas for numerical and tabular data work.
- Matplotlib and Seaborn for visualization.
- SciPy for statistical and probability tools.
- SQL for relational data querying.
- scikit-learn for classical machine learning and evaluation.
- Git and GitHub for version control and portfolio development.

PORTFOLIO PROJECT IDEAS

- Student marks analyzer with validation and file output.
- Exploratory data analysis project with cleaning, charts and written insights.
- SQL analysis project using joins and grouped summaries.
- Regression project with a baseline and error analysis.
- Classification project with precision, recall, F1 and confusion matrix.
- Clustering project with scaled features and careful interpretation.
- End-to-end capstone with a reproducible notebook, README, source attribution and limitations.

PROFESSIONAL LEARNING PRACTICES

- Write code yourself instead of only copying examples.
- Predict outputs before running code, then explain differences.
- Keep projects reproducible with clear setup instructions.

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- Document dataset sources, assumptions and limitations.
 - Review earlier concepts weekly and revisit weak areas.
 - Publish projects only when you can explain the methods and results.

EXPECTED OUTCOME

Completing this roadmap can establish a foundation and produce portfolio projects, but it does not guarantee job readiness in exactly 60 days. Progress depends on practice, prior knowledge, feedback, and project depth. Continue building, reviewing, and explaining your work beyond this schedule.

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